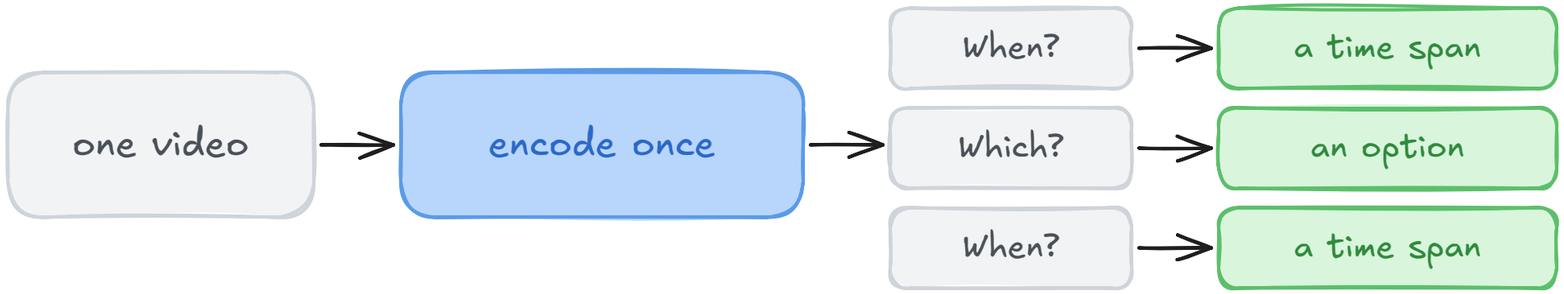
PAPER CLUB · OCTOBER 2026

VideoJev

System One Model for Video Understanding and Interaction

Yuhan Zhu, Xiangyu Zeng, Qingyi Si, … , Jiaqi Wang, Nan Duan, Limin Wang (18 authors)
Nanjing University (MCG-NJU), JD.com and six more universities

arXiv:2610.18615 (13 Oct 2026) · github.com/MCG-NJU/VideoJev · models: VideoJev-4B, VideoJev-35B-A3B



scores, not generated text:
one forward pass for all questions

A video model that answers by scoring candidates instead of writing answers.

Slides drawn in Excalidraw; every number is from the paper (see references/digest.md)

The paper in one slide

Problem

Video MLLMs answer by writing text token by token, even when the answer is a time span or 'B'.

- slow: 'Thinking' takes 145 s per question (Qwen3.5-4B)
- output must be parsed
- an agent asks many questions about the same video

Idea

A System One head on an MLLM: score candidates in one pass.

- grounding: score every short time interval → pick a span
- QA: score every option → softmax
- video encoded once, many questions in one forward pass

Evidence

Grounding, mean mIoU:

- 4B: 58.6 → 65.3 (+6.7)
- 35B-A3B: 56.9 → 67.7 (+10.8)
- 2.8–3.1× faster than backbone

Video QA, 4-benchmark mean:

- about equal to 'Thinking' (74.3 vs 74.9; 79.5 vs 78.8)
- 100–129× faster

Many video questions do not need a written answer, only a decision.

VideoJev (arXiv:2610.18615): abstract, Tables 1 and 2

Glossary

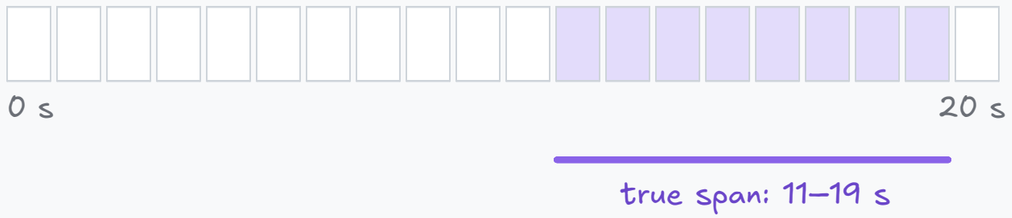
term	meaning
System One model	fast, one-pass model that returns typed decisions with probabilities (after Kahneman)
Jev	TypeSafe AI's System One model for text; VideoJev is a video version
temporal grounding	find when an event happens in a video: output a start and end time
mIoU · R1@0.7	mean overlap of predicted and true span · share of queries with overlap ≥ 0.7
multiple-choice video QA	a question about a video with 4–5 answer options; metric: accuracy
Instruct · Thinking	an MLLM answering directly · the same model reasoning in text first
prefill · KV cache	processing the prompt (here the video) once · stored attention states to reuse
Cached inference	prefill the video once; each new question continues from a copy of the cache
Joint inference	put all questions after the video and score them in one forward pass
35B-A3B	mixture-of-experts model: 35B parameters in total, about 3B used per token
ECE	expected calibration error: gap between confidence and actual accuracy

VideoJev (arXiv:2610.18615), §1, §2.1, §3.5; TypeSafe AI, 'Introducing System One Models & Jev'

Two everyday video questions: When? and Which?

Temporal grounding: When?

"When is chocolate added to the waffles?"



Output: a start and an end time.
Metrics: mIoU (mean overlap with the true span), R1@0.7 (overlap at least 0.7).
Benchmarks: Charades, ActivityNet, QVHighlights.

Multiple-choice video QA: Which?

"What is drizzled onto the waffles?"

- **A. Chocolate**
- B. Jam
- C. Cream

Output: one option.
Metric: accuracy.
Benchmarks: LVBench, Video-MME, LongVideoBench, MLVU (videos up to hours).

Both answers are tiny: two numbers, or one letter.

VideoJev (arXiv:2610.18615), Figure 2 (example redrawn) and §3.1

Today: video MLLMs write their answers

Even a time span or a letter is generated token by token

grounding output

```
"The event happens from 11.0 to 19.0 seconds."
```

QA output, Thinking mode

```
<think> The person pours something dark… it looks
like chocolate sauce… </think>  Answer: A
```

What this costs

- time per token, many tokens
- text must be parsed; invalid answers count as wrong
- no score for 'how sure'
- each new question decodes again

Video QA, latency per question on LVBench and mean accuracy (4 benchmarks)

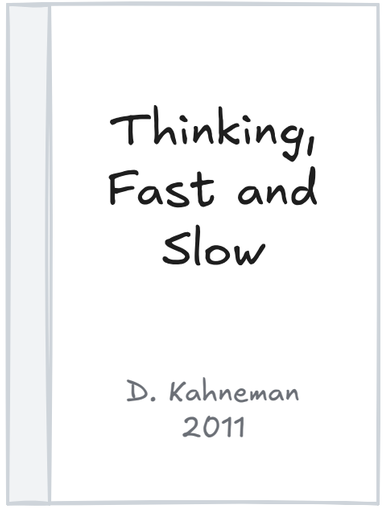
model	latency	accuracy
Qwen3.5-4B Instruct	1.36 s	66.3
Qwen3.5-4B Thinking	144.9 s	74.9

Thinking buys +8.6 points for about 107× the time.

VideoJev (arXiv:2610.18615), Table 2 (H200 GPU, batch size 1, model forward time); generated texts are illustrative

Where the name comes from: Thinking, Fast and Slow

Daniel Kahneman, 2011: two modes of thought



System 1 · fast

instinctive, emotional, automatic

2 + 2 = ? → 4 instantly, no effort

also: reading a face, a short word, noticing a sudden sound

System 2 · slow

deliberative, logical, effortful

17 × 24 = ? → 408 step by step

also: filling in a tax form, checking a logical argument

The same split in AI models

System 1	System One models (Jev, VideoJev): one forward pass → a typed decision with probabilities
System 2	LLMs that write free text; 'Thinking' mode: reason step by step, then answer

Kahneman: we use System 1 most of the time and call System 2 when it is needed. VideoJev aims for the same split.

D. Kahneman, Thinking, Fast and Slow (2011); 17 × 24 is Kahneman's own example. Book drawn generically, not the cover

The System One idea (Jev)

Fast typed judgments instead of slow free text

System 2: slow, deliberate

a chat LLM: writes free text token by token, can reason, plan, explain; output must be parsed by software

System 1: fast, intuitive

a System One model: returns a typed decision (a choice, a yes/no, a score) with probabilities, in one pass

Jev (TypeSafe AI)

- typed probabilistic judgments that software can use directly
- trained with RL for calibrated probabilities
- 'Jev' after the economist Jevons: cheaper decisions → many more uses
- image versions: Visual Jev, PixelJev → this paper: video

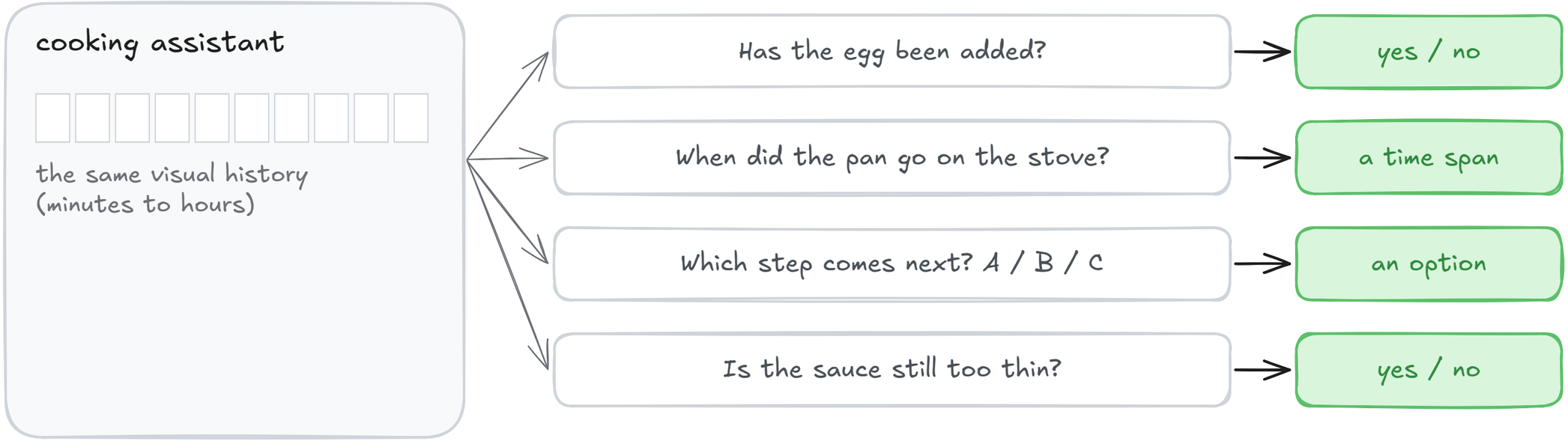
The paper's premise

"Structured visual judgments can support open-ended behavior without requiring open-ended generation at every step."

VideoJev (arXiv:2610.18615), §1 and §4; TypeSafe AI blog (vendor description of Jev)

Why video makes this harder

An agent asks many questions about the same video, over time



Requirement 1: time

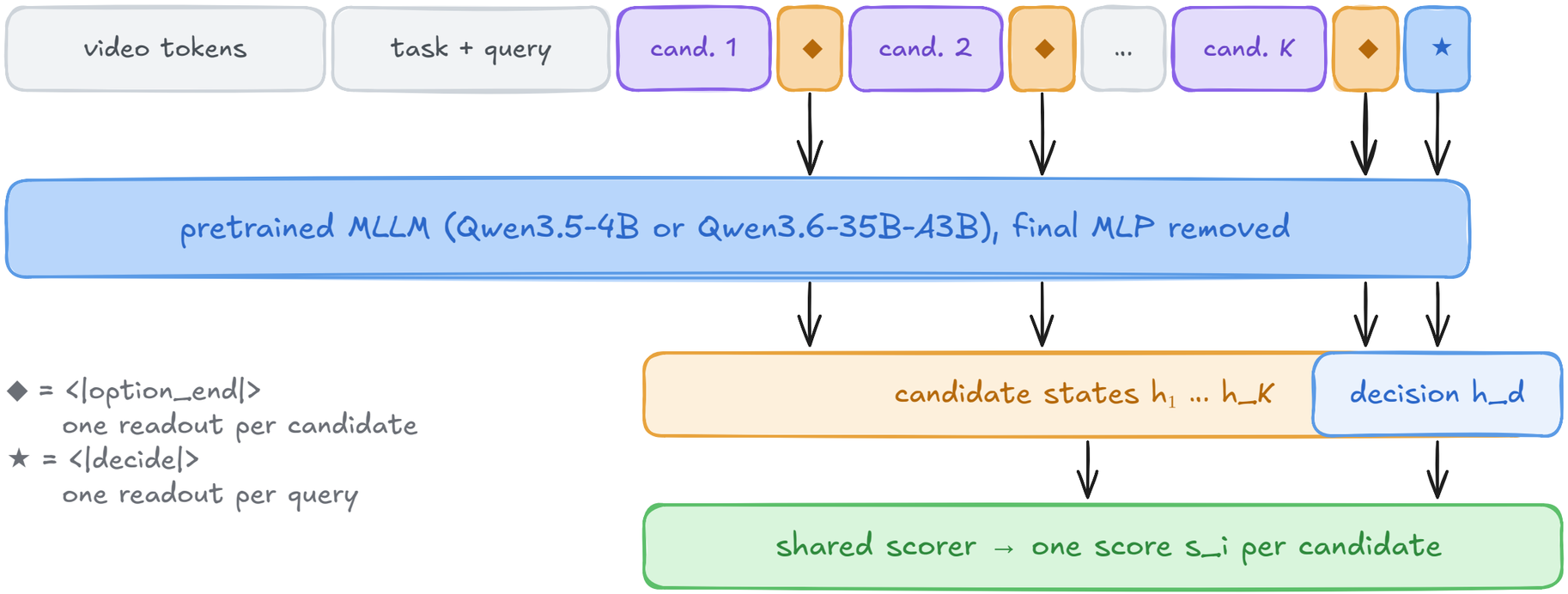
decisions must respect temporal structure
(a span, not a single frame)

Requirement 2: reuse

encoding the video is the expensive part;
every extra question should be cheap

VideoJev (arXiv:2610.18615), §1 (cooking example); questions are illustrative

Architecture: candidates in the prompt, scores at special tokens



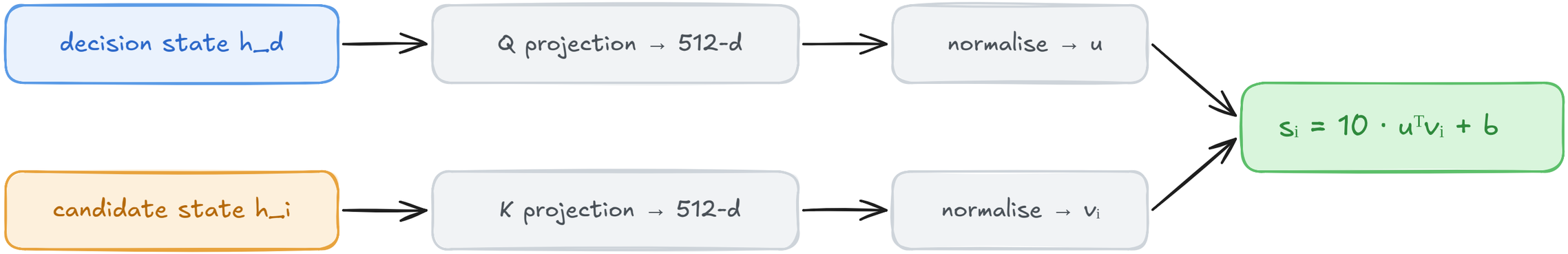
Why it works with one causal pass

each candidate readout sees the video, the query and earlier candidates; <|decide|> sees the whole list

Small changes to the MLLM

two new tokens (initialised from <|im_end|>), no LM head, no final MLP; the vision encoder stays frozen

The shared scorer: a dot product between question and candidate



- cosine similarity between 'what is asked' and 'this candidate', scaled by 10, plus a learned bias b
- the same head scores time intervals and text options; only the output rule differs:

grounding: sigmoid, one per interval

$p_i = \sigma(s_i)$: is this interval inside the event?

QA: softmax over the options

$p_i = \text{softmax}(s)_i$: which option is right?

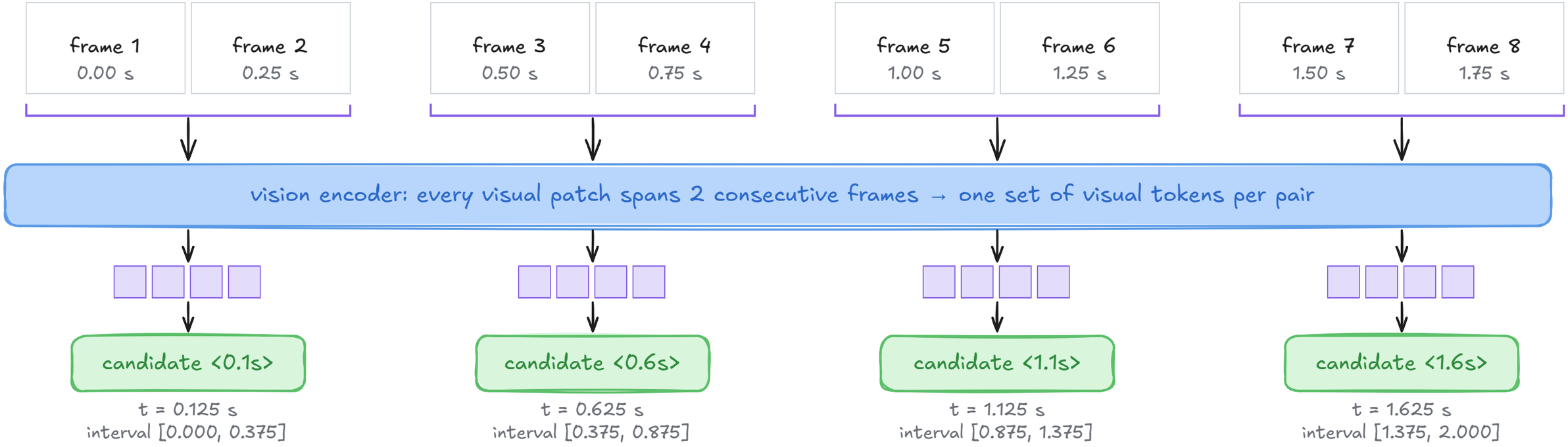
Projections start from small random weights (std 0.005), biases at zero. Why drop the final MLP? Pre-MLP features are known to carry semantic similarity (FreeRet); the ablation slide shows no loss.

VideoJev (arXiv:2610.18615), Eq. (1), §2.1 and §2.4

Why one candidate = two frames

The backbone compresses video twofold in time

frames sampled at 4 fps (as in evaluation)



Why not one candidate per frame?

frame 1 and frame 2 share their visual tokens:
the LLM never sees them apart, so a single-frame
candidate would have nothing of its own to point to

Consequences

one candidate = 2 / fps seconds: 0.5 s at 4 fps, 1 s at 2 fps
boundaries fall halfway between neighbouring pairs
a 2-minute video at 4 fps → 480 frames → 240 candidates

VideoJev (arXiv:2610.18615), §2.2 and Figure 2 ("The backbone's twofold temporal compression groups frames into pairs."); video of 2.0 s for illustration

Grounding as many small yes/no decisions

One candidate per frame pair (previous slide): is this interval inside the event?

candidates: timestamp text of each frame pair (copied from the video prefix); here 2 fps, so 1 s per candidate



label y_i = 1 if the pair's middle time lies inside the annotated event (here 11–19 s)

Loss 1: binary cross-entropy

each interval judged on its own, multi-label (several can be positive):

L_{BCE} = mean over i of $BCE(p_i, y_i)$

Loss 2: soft IoU (stage 2, weight 2)

rewards covering the whole event, penalises probability outside it, weighted by duration d_i:

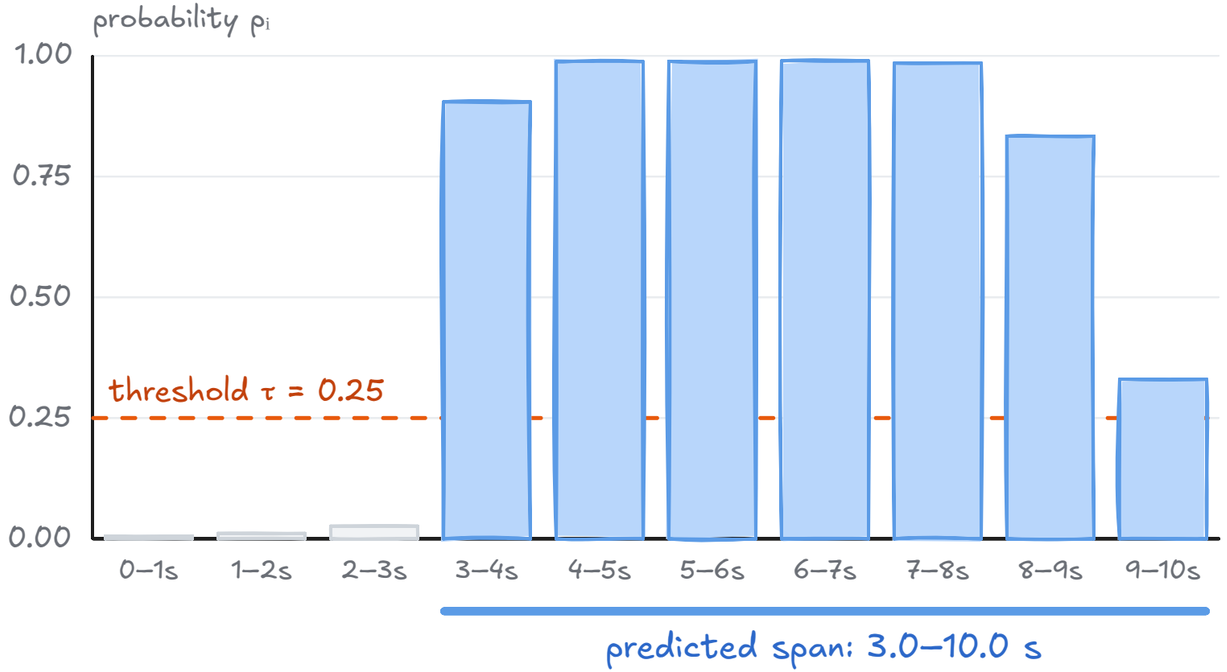
$$L_{IoU} = 1 - \Sigma\, d_i p_i y_i \,/\, \Sigma\, d_i [y_i + p_i(1-y_i)]$$

Soft IoU matters: without it, mean grounding mIoU drops from 65.3 to 60.7.

VideoJev (arXiv:2610.18615), §2.2, Eq. (2)–(3) and Table 6; 20 intervals drawn for illustration

From interval scores to one span

Real output of VideoJev-4B on a 10-second clip (README demo)



Decoding rule

1. shift each logit by the threshold:
   $a_i = s_i - \text{logit}(\tau)$
   ($a_i > 0$ exactly when $p_i > \tau$)
2. pick the contiguous range with the largest sum of a_i (Kadane, $O(K)$)
3. its interval boundaries are the answer; if all $a_i < 0$, take the best single interval

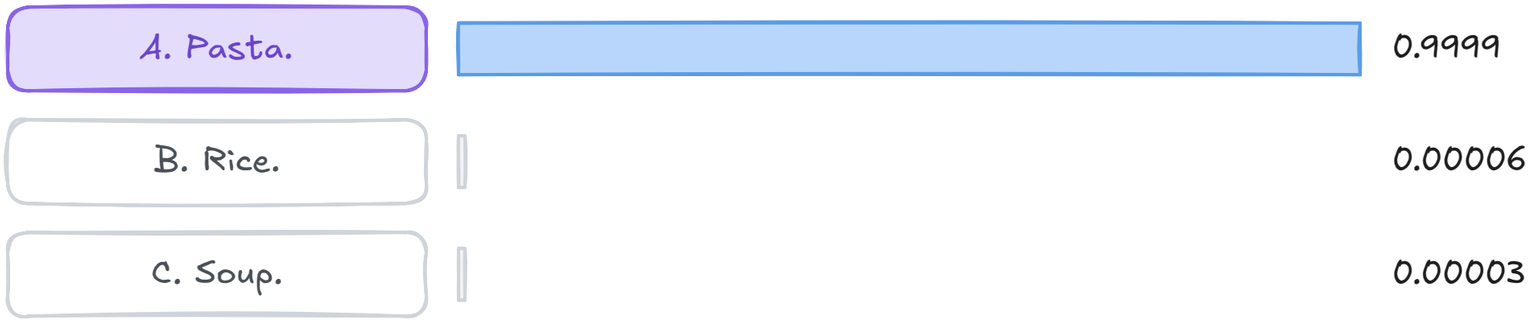
shifted logits a_i: `-4.0 -3.4 -2.5 +3.3 +5.5 +5.6 +5.7 +5.3 +2.7 +0.4`

The last interval ($p = 0.33$) is still above τ, so the span runs to the end of the clip. $\tau = 0.25$ is fixed for all experiments and both model sizes.

github.com/MCG-NJU/VideoJev README (query: 'The woman in green eats pasta.'); VideoJev (arXiv:2610.18615), Eq. (4)

Video QA as choosing among scored options

"What is the woman in green eating?" (README demo, VideoJev-4B)



Training

exactly one option is correct:
softmax over the option scores,
cross-entropy on the right one

~240K questions over 19K videos
(internal data, 4–5 options each)

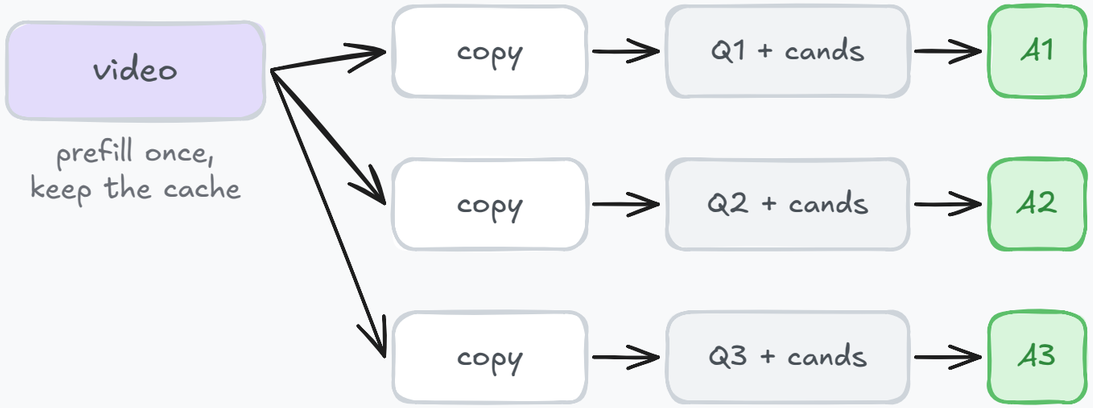
Avoiding a position shortcut

options are shuffled in training
(reproducibly, independent of the
answer), so 'A' is not always right;
questions that refer to positions
or earlier turns keep their order

github.com/MCG-NJU/VideoJev README; VideoJev (arXiv:2610.18615), §2.3, Eq. (5) and §2.4

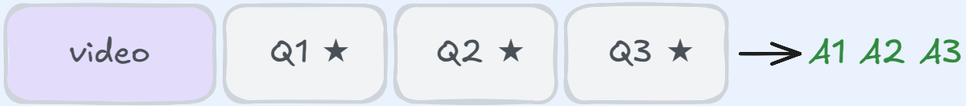
Encode the video once: Cached and Joint inference

Cached inference: questions arrive one by one



one short forward per question

Joint inference: questions known together



one causal forward pass for all; each question has its own <|decide|> readout (★)

later questions can attend to earlier questions and options, but no answers are inserted (standard causal attention, no custom mask)

Training uses the same layout: several questions per video share one prefix.

Generative baselines can reuse the video prefix too, but must still decode each answer token by token; only single-token answers (a letter) can be read jointly.

Training recipe

stage	trained	videos	grounding loss	QA loss	updates
1 · head alignment	scoring head only	30%	BCE	cross-entropy	160
2 · joint adaptation	language backbone, new tokens, head	70%	BCE + 2 × soft IoU	cross-entropy	375

Data

grounding: TimeLens-100K + TimeLens2-93K (intervals turned into per-candidate labels)

QA: ~240K multiple-choice questions over 19K videos, 4 min to ~2 h (internal, not released)

Setup

backbones: Qwen3.5-4B, Qwen3.6-35B-A3B
vision encoder frozen throughout
64 H200 GPUs, 2 fps, up to 144K tokens
head LR 3e-4 / 1e-4; backbone LR 5e-6 (4B), 2e-6 (35B-A3B); one epoch per stage

Why two stages? Like projector alignment in LLaVA: give the new head a stable feature space first. It is also cheaper.

VideoJev (arXiv:2610.18615), §2.4 and §3.1

Temporal grounding: best scores, lowest latency

mIoU per benchmark (TimeLens-refined annotations); latency per query on full Charades

model	latency	Charades	ActivityNet	QVHighlights	mean
GPT-5	–	40.5	42.9	56.8	46.7
Gemini-2.5-Pro	–	52.8	58.1	70.4	60.4
Qwen3.5-4B (backbone)	0.763 s	52.3	55.6	67.9	58.6
TimeLens2-4B (best 4B)	0.646 s	57.7	59.0	69.3	62.0
VideoJev-4B	0.276 s	60.1	62.8	73.2	65.3
Qwen3.6-35B-A3B (backbone)	1.054 s	47.3	56.0	67.5	56.9
VideoJev-35B-A3B	0.336 s	63.7	65.5	74.0	67.7

+6.7 and +10.8 mIoU over the backbones, 2.8–3.1× faster, and above Gemini-2.5-Pro.
Stricter overlap improves most: R1@0.7 vs TimeLens2-4B is +2.8 to +4.8 points per benchmark.

VideoJev (arXiv:2610.18615), Table 1 (selected rows; all open models in the table are below VideoJev)

Video QA: as good as Thinking, about 100× faster

Accuracy (MLVU: M-Avg); latency per question on full LVBench

model	latency	LVBench	Video-MME	LongVideoBench	MLVU	mean
Qwen3.5-4B Instruct	1.4 s	60.0	70.6	61.2	73.3	66.3
Qwen3.5-4B Thinking	144.9 s	66.4	76.9	73.6	82.8	74.9
VideoJev-4B	1.1 s	67.9	75.0	73.6	80.6	74.3
Qwen3.6-35B-A3B Instruct	2.4 s	66.0	77.1	72.4	79.5	73.8
Qwen3.6-35B-A3B Thinking	149.8 s	71.4	82.5	74.9	86.2	78.8
VideoJev-35B-A3B	1.5 s	73.4	81.8	77.1	85.9	79.5

vs Instruct backbones

+8.0 (4B) and +5.7 (35B-A3B) mean points, at similar latency

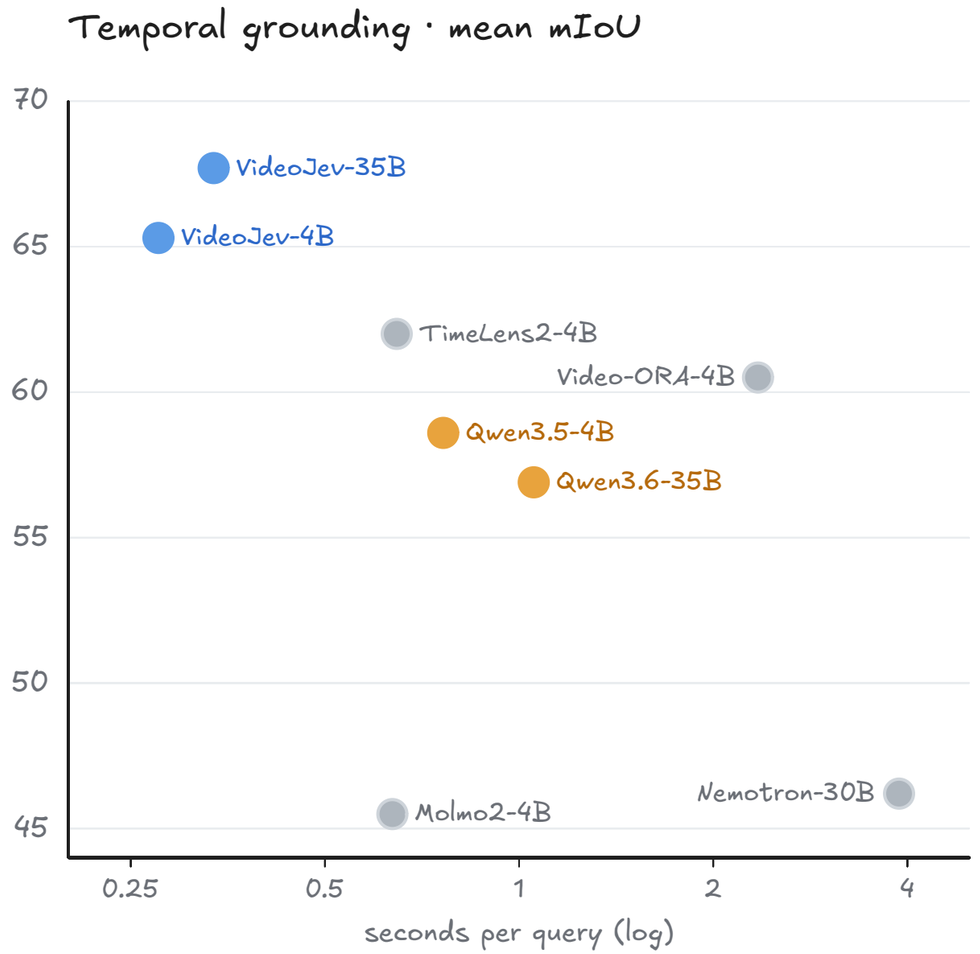
vs Thinking

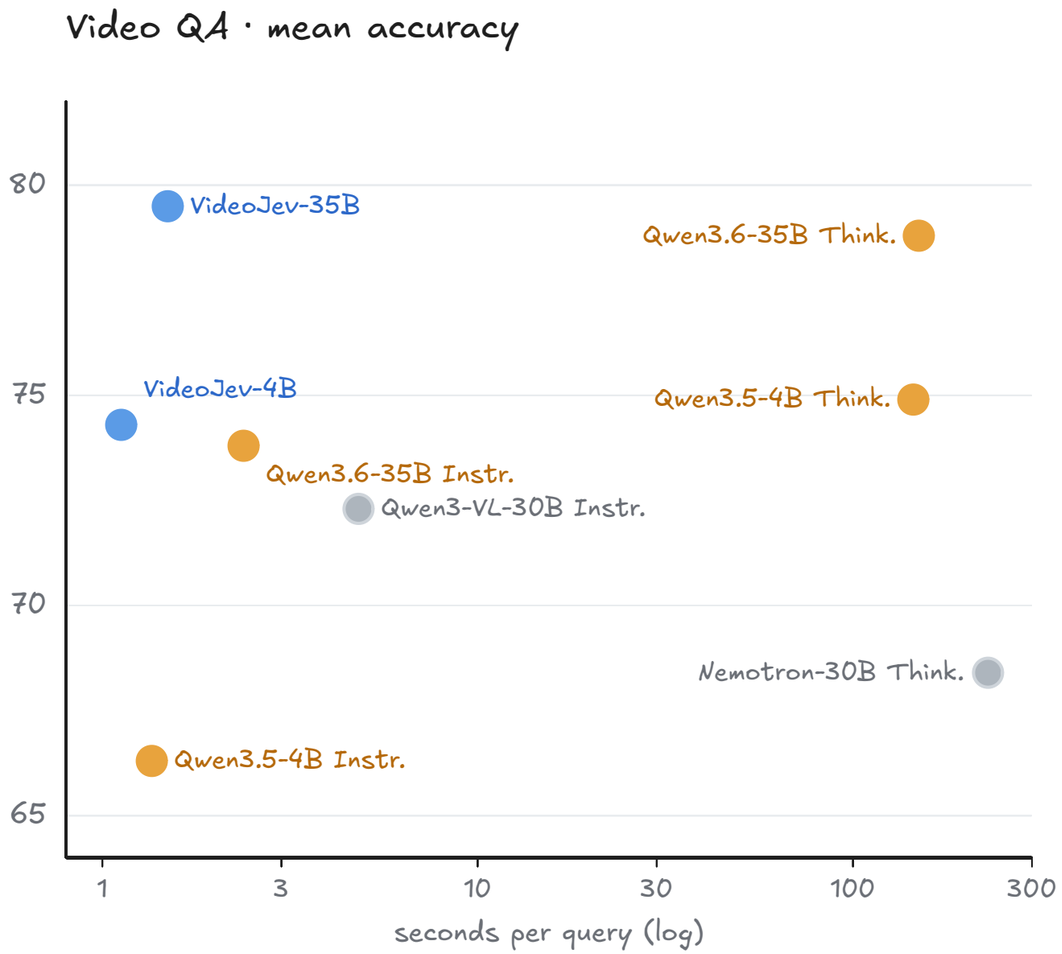
−0.6 (4B) and +0.7 (35B-A3B) mean points, 129× and 100× faster; Thinking still wins on Video-MME and MLVU

VideoJev (arXiv:2610.18615), Table 2 (frames only, no subtitles or audio)

Quality against latency

Upper left is better; the x-axis is logarithmic

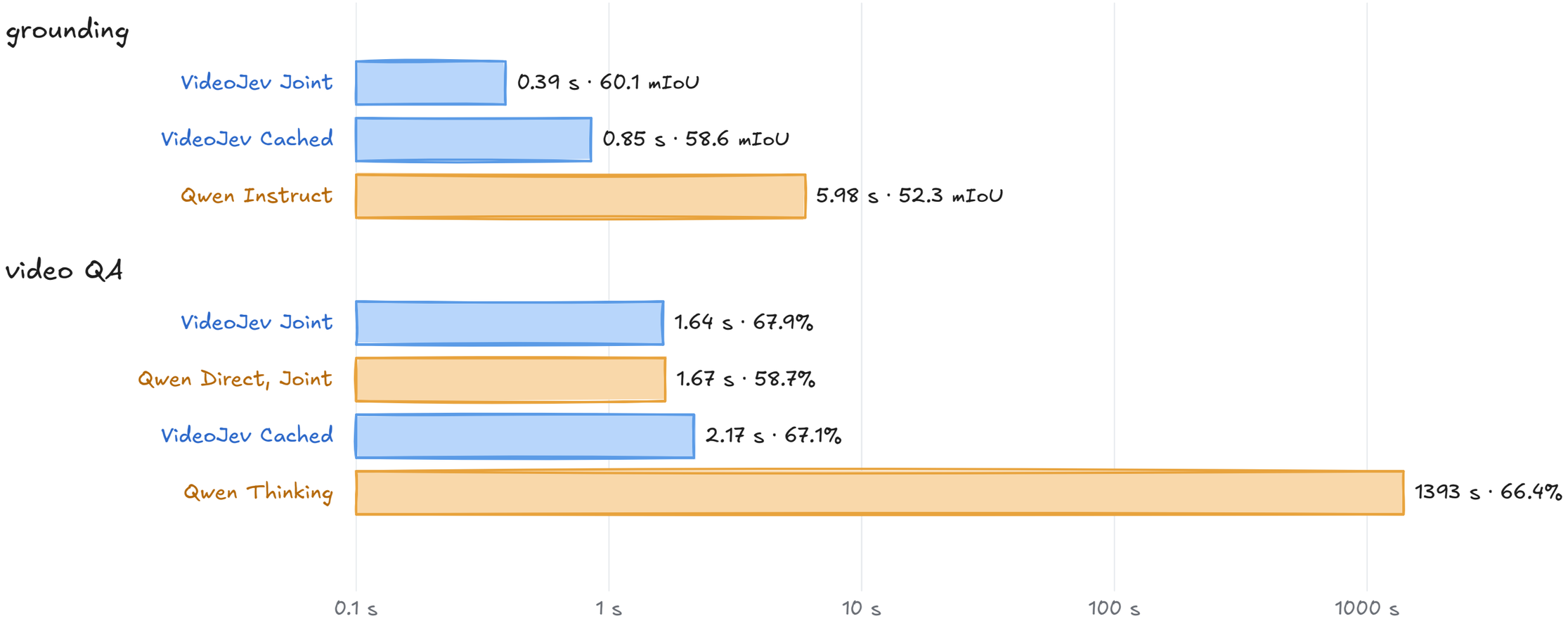




VideoJev (arXiv:2610.18615), Tables 1–2 and Figure 1 (selected models redrawn)

Ten questions about one video

Total model time for 10 queries per video, VideoJev-4B vs Qwen3.5-4B (log scale)



Joint inference: going from 1 to 10 questions adds only 55 ms (grounding) and 31 ms (QA).

VideoJev (arXiv:2610.18615), §3.5 and Figure 3 (40 videos per task, H200, batch 1); quality from the full benchmarks, Cached QA from Table 4

Is it the architecture or the training data?

Same data, same single-stage schedule: scoring head vs ordinary autoregressive fine-tuning

model	Charades	ActivityNet	QVHighl.	mean	LVBench	Video-MME	LongVB	QA mean
VideoJev-4B (single stage)	60.6	62.8	72.9	65.4	67.7	74.9	73.5	74.7
Qwen3.5-4B, autoregressive SFT	61.0	61.3	70.0	64.1	66.9	74.5	71.4	72.8

Autoregressive model: generates the interval as JSON in seconds, or the option letter; same videos, same frozen encoder.

Scoring wins on average

+1.3 grounding mIoU and +1.9 QA points with identical supervision: the readout fits the decision better

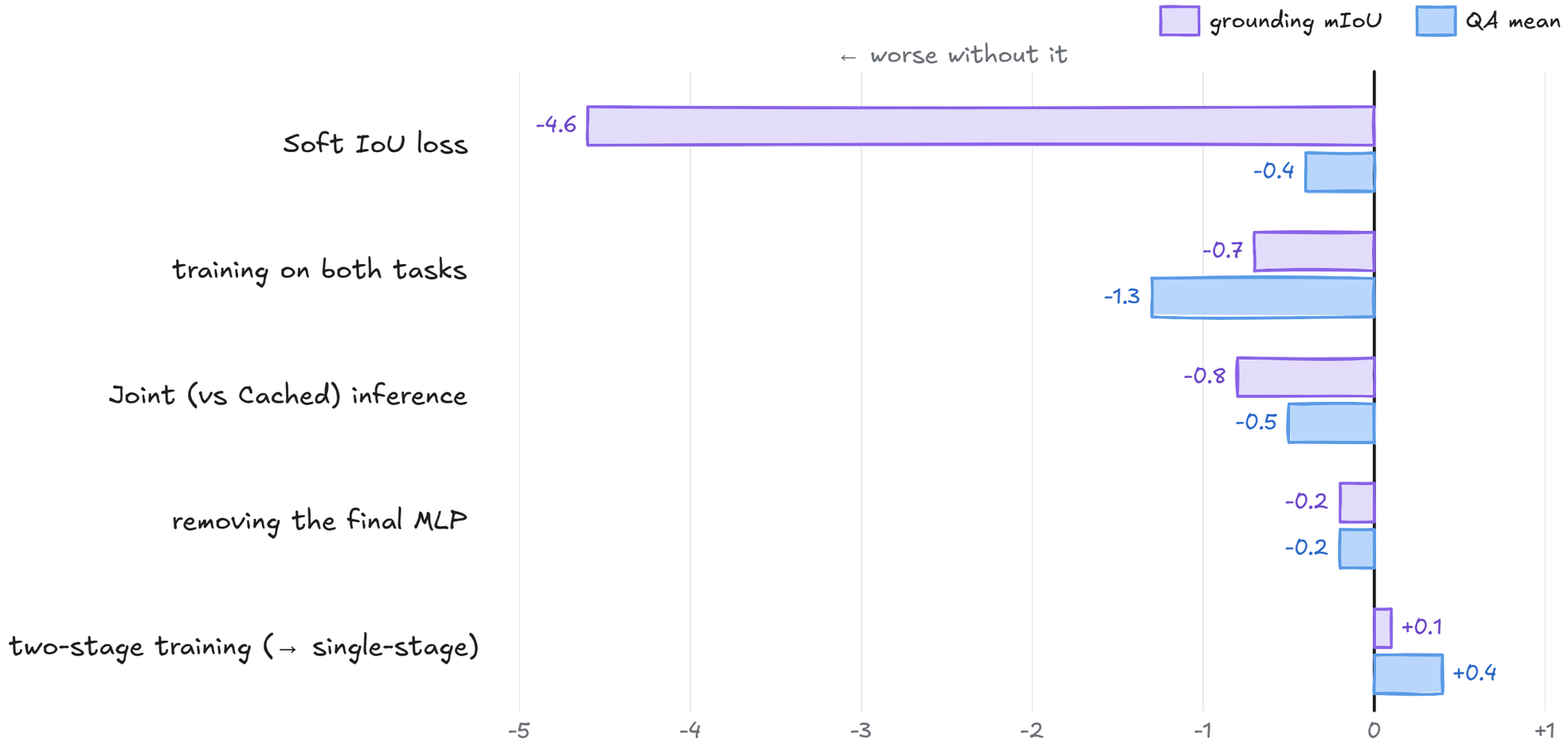
But the gap is modest

most of the gain over the backbone comes from fine-tuning itself; autoregressive SFT is even better on Charades (+0.4)

VideoJev (arXiv:2610.18615), Table 3 and App. A.3 (the MLVU cell of the autoregressive row is empty in the HTML table)

Ablations: what each design choice is worth

Change in mean score when the choice is removed (VideoJev-4B)

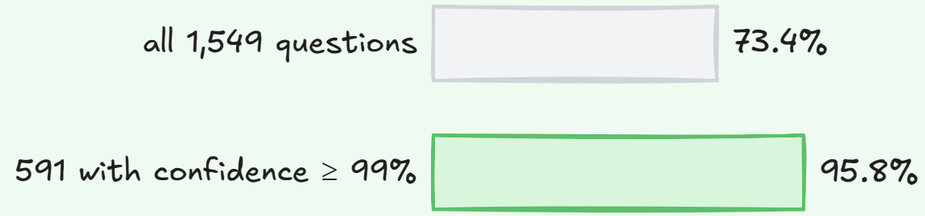


Soft IoU is the big one. Single-stage training is slightly better (+0.4 QA, mostly MLVU); two stages are kept because they are cheaper (backbone updated on 70% of the videos).

VideoJev (arXiv:2610.18615), Tables 4–8

Confidence: useful for ranking, not yet calibrated

High confidence → high accuracy



VideoJev-35B-A3B on LVBench

But probabilities are overconfident

expected calibration error, mean of 4 QA benchmarks:

8.65 points (4B)
8.13 points (35B-A3B)

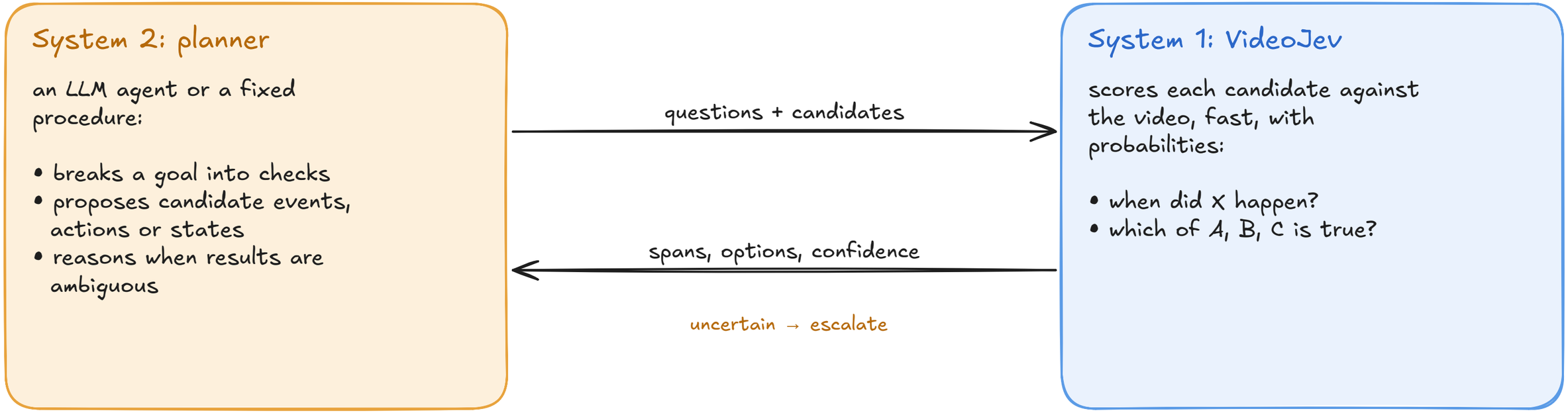
no calibration-specific training

A difference from Jev

Jev is trained with reinforcement learning to output calibrated probabilities; VideoJev is not.
The authors suggest temperature scaling on held-out data or calibration-aware training.

VideoJev (arXiv:2610.18615), §3.5 and Figure 4

Where this fits: System 2 asks, System 1 checks



"System Two organizes the decision space, while System One evaluates it against visual evidence."

Applications named: procedural monitoring, interactive assistance, embodied agents, computer use. Next steps: more task types (action recognition, event detection), streaming input, better calibration.

VideoJev (arXiv:2610.18615), §5 (future work; not evaluated in the paper)

Reading it critically

Strong

- one head unifies two task types; simple, reusable design
- careful latency protocol: generative baselines also reuse the video prefix
- controlled comparison with autoregressive SFT on the same data
- models and code released

Weak or unclear

- closed set: someone must supply the candidates; one contiguous span per query only
- QA training data is internal; gains partly from data (scoring adds +1.3 / +1.9 over autoregressive SFT)
- latency excludes video decoding and preprocessing, which also cost time in real systems
- Joint inference: later questions see earlier ones; order effects are not studied
- overconfident probabilities (ECE ≈ 8 points)
- 'System One' comes from a vendor blog

Takeaways

1 · Score, don't write

For spans and choices, scoring beats generating: +6.7 / +10.8 mIoU over the backbones, +1.3 over autoregressive fine-tuning on the same data.

2 · Encode the video once

Joint inference: ten questions cost almost the same as one (+55 ms grounding, +31 ms QA).

3 · Close to Thinking without thinking

Video QA within 0.6 points of, or above, Thinking mode, at about 1/100 of the time.

4 · Training choices matter

Soft IoU is worth +4.6 mIoU; training grounding and QA together helps both tasks (+0.7 / +1.3).

5 · Confidence is a bonus, not yet reliable

High confidence predicts correctness, but the probabilities are overconfident (ECE ≈ 8 points).

arxiv.org/abs/2610.18615 · github.com/MCG-NJU/VideoJev